CASE STUDY REPORT

Advantages of Using AI in Government and the Public Sector

U.S. Treasury Payment Integrity and an AI Resilience Proposal for Houston

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Prepared as a report examining measurable benefits, implementation challenges, and responsible innovation in public-sector AI.

Introduction

Artificial intelligence is becoming an important tool for governments because public agencies must serve large populations, manage limited resources, detect complex risks, and make decisions using enormous amounts of data. Unlike a private company, however, a government agency cannot judge success only by speed or cost savings. Public-sector AI must also protect constitutional rights, maintain transparency, provide equal treatment, and preserve public trust. When those conditions are met, AI can improve service delivery, help employees focus on difficult cases, identify fraud, forecast emergencies, and support evidence-based policy decisions.

The greatest advantage of AI in government is its ability to find useful patterns at a scale that would overwhelm human analysts. Machine-learning models can review millions of transactions, sensor readings, documents, or service requests and flag the small number that need attention. Natural-language processing can help agencies organize public comments, answer routine questions, and search large collections of regulations. Predictive analytics can estimate where infrastructure failures, disease outbreaks, traffic congestion, or disaster impacts are most likely to occur. These tools do not eliminate the need for public employees; instead, they can give employees better information and allow them to spend more time on judgment, communication, and direct assistance.

This report examines the U.S. Department of the Treasury's use of machine-learning methods to prevent and recover fraudulent or improper federal payments. The case is valuable because it includes a clearly defined public need, documented AI techniques, and measurable results. The report then proposes a locally focused application: an AI-powered resilience and resource-allocation system for the Houston region that would help public agencies prepare for flooding, extreme heat, and related infrastructure disruptions.

Case Study Analysis: U.S. Treasury Payment Integrity

Problem or Need Addressed

The federal government sends benefits, grants, tax refunds, vendor payments, and other funds to millions of people and organizations. Treasury serves as the central disbursing agency and reports that it securely sends about 1.4 billion payments worth more than $6.9 trillion to over 100 million people each year (U.S. Department of the Treasury, 2024). At that scale, even a small error rate can produce major losses. The growth of stolen identities, altered checks, synthetic identities, compromised accounts, and organized fraud after the COVID-19 pandemic increased the difficulty of distinguishing legitimate payments from suspicious ones.

Traditional fraud controls depend heavily on fixed rules, manual reviews, and known watch lists. Those methods remain useful, but criminals can adjust their behavior once a rule becomes predictable. Human reviewers also cannot examine every payment in depth. Treasury therefore needed a system that could combine data from many sources, score risk quickly, and identify unusual patterns early enough to stop or recover payments without unnecessarily delaying legitimate recipients.

AI Tools and Techniques Implemented

Treasury's Office of Payment Integrity, located within the Bureau of the Fiscal Service, expanded its technology- and data-driven fraud controls. The program uses risk-based screening and machine-learning models to prioritize transactions that display characteristics associated with fraud or improper payment. Machine learning is especially useful because it can evaluate combinations of variables rather than relying on a single rule. A transaction may appear normal when each field is viewed separately, yet become suspicious when payment timing, account history, identity data, check characteristics, and connections to prior fraud are considered together.

The AI system supports, rather than replaces, investigators and program officials. High-risk cases can be routed for additional verification, while lower-risk payments continue through normal processing. Treasury also expanded data-sharing partnerships, including access for state unemployment agencies to federal payment-integrity data through the Unemployment Insurance Integrity Data Hub. This shared-service approach creates an additional public-sector advantage: one agency can build specialized analytical capability and make it available to many programs instead of requiring every agency to develop an independent fraud platform.

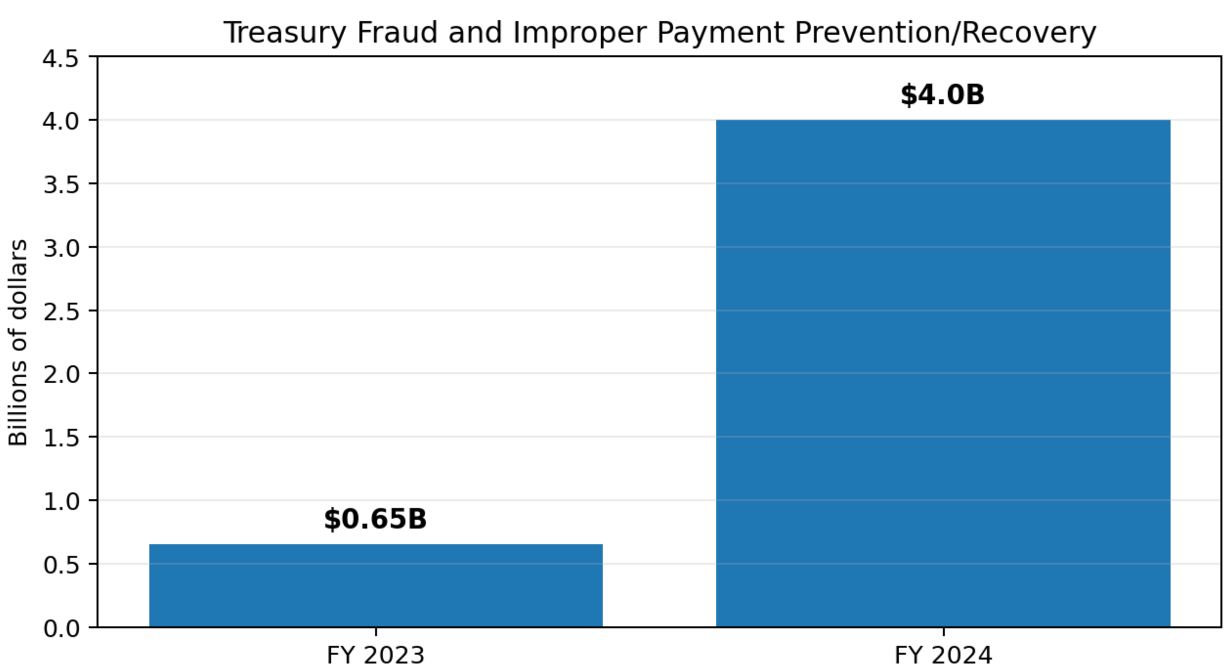

Figure 1. Reported Treasury fraud and improper-payment prevention/recovery increased from $652.7 million in FY2023 to more than $4 billion in FY2024. Source: U.S. Department of the Treasury (2024).

Outcomes and Benefits Achieved

Treasury reported that its enhanced processes prevented or recovered more than $4 billion during fiscal year 2024, compared with $652.7 million in fiscal year 2023. The FY2024 total included approximately $500 million from expanded risk-based screening, $2.5 billion from identifying and prioritizing high-risk transactions, $1 billion recovered through faster identification of Treasury check fraud using machine-learning AI, and $180 million from improvements to payment-processing schedules (U.S. Department of the Treasury, 2024). The reported increase is more than six times the prior-year amount, demonstrating why AI can be valuable when it is applied to a high-volume, data-rich government function.

The financial benefit is only one outcome. Faster detection reduces the time available for stolen funds to be moved or laundered. Better prioritization helps investigators concentrate on the cases with the highest expected risk instead of reviewing transactions in random order. Shared analytical services can improve consistency across agencies, and prevented losses protect funds intended for citizens, public programs, and legitimate vendors. The program also illustrates how AI can improve stewardship: government can provide services at scale while reducing waste and protecting taxpayer resources.

Treasury capability	Reported FY2024 result	Public-sector advantage
Expanded risk-based screening	$500 million prevented	Screens transactions before payment
High-risk transaction prioritization	$2.5 billion prevented	Directs review resources to likely fraud
Machine-learning check-fraud detection	$1 billion recovered	Accelerates detection of suspicious checks
Payment schedule efficiencies	$180 million prevented	Creates time for verification and intervention

Table 1. Major components of Treasury's FY2024 payment-integrity results.

Challenges and Limitations

Despite the strong results, fraud-detection AI creates important risks. First, false positives can delay lawful payments to people who may depend on benefits or refunds for basic needs. A model trained on incomplete or historically biased data may assign higher risk to certain communities, payment types, or geographic areas without a valid reason. Government agencies must therefore test error rates across groups, document the variables used, and provide a timely human review and appeal process.

Second, fraud models are difficult to operate in a fully transparent manner. Publishing every detection rule could help criminals evade the system, but excessive secrecy can make it impossible for oversight bodies or affected citizens to understand why a payment was stopped. A responsible balance would protect sensitive detection details while still documenting model purpose, data sources, performance measures, audit results, and the process for correcting errors.

Third, fraud patterns change. A model that performs well today may become less accurate as criminals adopt new methods or as government programs change. Continuous monitoring is therefore essential. The Government Accountability Office recommends that public-sector AI accountability address governance, data, performance, and monitoring, while the National Institute of Standards and Technology emphasizes managing risks to individuals, organizations, and society throughout the AI life cycle (GAO, 2021; NIST, 2023). Treasury must also protect sensitive personal and financial data, restrict employee access, maintain cybersecurity controls, and ensure that vendors do not gain inappropriate control over public decision-making.

Innovative Proposal: Houston AI Resilience and Resource Network

The Public-Sector Issue

The Houston metropolitan area faces overlapping threats from flooding, hurricanes, extreme heat, power outages, traffic disruption, and rapid population growth. Public agencies already use weather forecasts, flood maps, emergency calls, infrastructure records, and demographic data, but these sources often exist in separate systems. During a fast-moving emergency, decision-makers must determine where to position rescue vehicles, cooling centers, generators, road crews, shelters, and public alerts. A delay of even an hour can increase harm, especially for older adults, people with disabilities, households without vehicles, and neighborhoods with limited access to health care or reliable power.

Proposed AI Application

I propose a Houston AI Resilience and Resource Network, a regional decision-support platform shared by the City of Houston, Harris County, emergency management offices, transit agencies, public works departments, health systems, and utility partners. The system would create a continuously updated digital representation of community conditions. It would combine rainfall and heat forecasts, bayou and drainage sensors, 311 and 911 trends, road closures, power-outage information, hospital capacity, shelter status, transit availability, and anonymized vulnerability indicators.

Machine-learning models would forecast likely impacts at the neighborhood level over the next several hours and days. An optimization engine would then recommend where to pre-position resources. For example, before an extreme-heat event, the system could identify neighborhoods with high forecast temperatures, low tree cover, older housing, limited vehicle access, and recent power instability. It could recommend opening mobile cooling stations, extending transit routes, staging water and generators, and sending multilingual alerts. Before a flood, it could estimate which intersections and facilities are most likely to become inaccessible and suggest alternate ambulance routes or locations for high-water vehicles.

The system should be designed as a human-in-the-loop tool. It would display confidence ranges and the evidence behind each recommendation rather than issuing automatic orders. Emergency managers would approve actions, record why they accepted or rejected recommendations, and use those decisions to improve future performance. Residents could access a simplified public dashboard showing verified hazards, open facilities, transit options, and the basis for major alerts. FEMA's Resilience Analysis and Planning Tool and OpenFEMA datasets demonstrate that government already maintains geospatial, hazard, infrastructure, and disaster data that could support this type of regional capability (FEMA, 2026a, 2026b).

1. Data	2. AI analysis	3. Human decision	4. Public action
Weather, sensors, roads, calls, shelters, outages	Neighborhood forecasts and resource optimization	Emergency managers review confidence and equity impacts	Crews, alerts, transit, shelters, cooling and rescue resources

Figure 2. Human-in-the-loop workflow for the proposed resilience network.

Expected Outcomes and Justification

The proposal is justified because emergency management is a resource-allocation problem under uncertainty. AI is well suited to combine rapidly changing data and compare thousands of possible deployment choices. Expected benefits include earlier warnings, shorter response times, better coordination across jurisdictions, reduced duplication of effort, and more equitable placement of services. The platform could also support long-term infrastructure planning by identifying repeated patterns, such as intersections that frequently flood, neighborhoods that experience recurring heat-related medical calls, or facilities that become isolated during storms.

Success should be measured with public metrics rather than claims about model accuracy alone. Useful measures would include warning lead time, emergency response time, percentage of residents within reasonable distance of a cooling center or shelter, reductions in repeated road closures, false-alarm rates, service distribution across neighborhoods, and after-action feedback from residents and first responders. A pilot could begin with one flood-prone watershed and one summer heat season before regional expansion.

Potential Challenges and Safeguards

The largest challenge is data governance. Emergency data can reveal sensitive information about health, location, disability, or household circumstances. The system should use the minimum necessary data, favor aggregated or anonymized records, enforce strict retention limits, and prohibit use for unrelated law-enforcement or immigration purposes. Cybersecurity is also critical because a compromised dashboard or manipulated sensor feed could direct resources to the wrong location.

Equity must be tested explicitly. Neighborhoods with fewer sensors or lower rates of 311 reporting may appear to have fewer needs even when the opposite is true. The city should conduct community reviews, publish impact assessments, monitor outcomes by area, and include non-digital reporting channels. Following GAO's accountability principles, the project should establish clear governance, validate data quality, evaluate performance, and continuously monitor results. Following the NIST AI Risk Management Framework, the city should document risks, define responsible officials, test the system under realistic failure conditions, and maintain manual fallback procedures.

Finally, procurement and workforce issues must be addressed. The region should avoid a contract that locks public agencies into a single vendor or prevents independent auditing. Data formats and model documentation should be portable. Public employees need training to understand what the system can and cannot predict. The goal is not to automate emergency authority; it is to give trained professionals a faster, broader, and more evidence-based view of changing conditions.

Conclusion

The Treasury case study shows that AI can produce substantial public value when it is matched to a specific operational problem and supported by reliable data, measurable goals, and human oversight. Treasury's machine-learning and risk-screening tools helped prevent or recover more than $4 billion in fraudulent and improper payments in fiscal year 2024. The program improved the government's ability to protect taxpayer funds while directing investigators toward the most important cases.

At the same time, the case demonstrates that public-sector AI cannot be evaluated only by savings. Agencies must protect privacy, measure false positives, explain decisions at an appropriate level, monitor changing performance, and allow people to challenge harmful errors. These requirements are especially important because government decisions can affect a person's income, safety, mobility, and access to essential services.

The proposed Houston AI Resilience and Resource Network applies the same basic advantage—finding patterns in large, changing datasets—to disaster preparedness and infrastructure coordination. With strong safeguards and human control, it could help local agencies act earlier, distribute resources more fairly, and reduce preventable harm from floods and extreme heat. The most successful government AI systems will not be the ones that replace public servants. They will be the ones that make public servants more informed, accountable, and effective.

References

Federal Emergency Management Agency. (2026a). Resilience Analysis and Planning Tool (RAPT). https://www.fema.gov/emergency-managers/practitioners/resilience-analysis-and-planning-tool

Federal Emergency Management Agency. (2026b). OpenFEMA. https://www.fema.gov/about/reports-and-data/openfema

National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). https://doi.org/10.6028/NIST.AI.100-1

U.S. Department of the Treasury. (2024, October 17). Treasury announces enhanced fraud detection processes, including machine learning AI, prevented and recovered over $4 billion in fiscal year 2024. https://home.treasury.gov/news/press-releases/jy2650

U.S. Government Accountability Office. (2021). Artificial intelligence: An accountability framework for federal agencies and other entities (GAO-21-519SP). https://www.gao.gov/products/gao-21-519sp

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